

# Test 1

Aditya

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## 1 Neural Networks

the neural network takes two inputs,  $x_1=1$  and  $x_2=2$ , weights are  $w_1=0.5$  and  $w_2=-0.3$  and a bias is  $b=0.1$  we will take the inputs and multiply them with weights, after that we will use the activation function on it

$$Z = (1)(0.5) + (2)(-0.3) + 0.1$$

$$z = 0.5 - 0.6 + 0.1$$

$$z = 0$$

bias is an extra value added to the weighted sum before the activation function. we can consider it as a part of the function which focuses its output in a specific direction

## 2 Inferencing

the number of parameters are 500 million, the stored format is fp16. in order to load the weights of the model weights would be

total memory = total numbers we have stored x amount of storage each number would take

fp16 = 500 million x 2 bytes = would be 1 billion bytes which would be 1GB of space

in inference the model only performs forward pass and makes predictions which requires less computations, whereas the training stage requires forward pass, calculating losses and updating gradient, these operations require more computation power